



Phelia Hoe Ying Qian

Contact



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Profile

An undergraduate at the Singapore University of Technology and Design (SUTD), pursuing a degree in Engineering Product Development with a specialization in Electrical Engineering. Nanyang Polytechnic graduate with a diploma in Aerospace Systems & Management. A motivated and proactive individual with strong initiative, communication, leadership, and customer-oriented skills.

Skills

- Leadership & Teamwork
- Adaptability & Efficiency
- Communication
- Project Management
- Problem Solving
- Electronics
- Prototyping
- Fabrication

Involvement

Executive member of SUTD Climber's Club (Publicity Director)

- Promoted the club through social media marketing to increase outreach and engagement
- Created and edited videos and posts to showcase club events and initiatives
- Organiser of Climbnival 2025
- Organised DSUTD 2025
- Promoted StuOrg showcase 2025
- Club representative for SUTD health & safety day 2025
- Organiser of Climbnival 2026

Experience

SUPER SKATE SQUAD (2024 - PRESENT)

Inline Skating Coach

- Taught children ranging from ages 3 to adults the fundamentals and advance techniques of skating
- Ensured the safety and well-being of all students during lessons
- Helped students develop confidence, discipline and achieve their fullest potential
- Applied diverse coaching methods to suit different learning styles and skill levels
- Instilled positive values in students

CHANGI AIRPORT GROUP (MAR 2023 - AUG 2023)

Changi Youth Ambassador (Student Intern)

- Assisted Duty Terminal Managers and ground handlers in smooth execution of daily airport operations
- Provided passenger assistance and front-line customer service, addressing enquiries and ensuring a positive travel experience
- Managed passenger queues at key airport areas to maintain efficient crowd flow
- Coordinated with airport staff to resolve on-ground issues promptly and maintain service standards

HIROLLER SINGAPORE (2020 - 2023)

Inline Skating Coach

- Taught children ranging from ages 4 to adults the fundamentals and advance techniques of skating
- Ensured the safety and well-being of all students during lessons
- Helped students develop confidence, discipline and achieve their fullest potential
- Applied diverse coaching methods to suit different learning styles and skill levels
- Instilled positive values in students

Marshaller

- Ensured safety of patrons when inline skating
- Provided guidance to patrons on proper skate usage
- Assisted patrons with gearing up ensuring equipment was properly worn
- Issued and maintained skating equipment

Education

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN (2024-PRESENT)

Bachelor of Engineering (Engineering Product Development)

2025

- Executive member of SUTD Climber's Club (2025-2026)

NANYANG POLYTECHNIC (2021-2024)

Diploma in Aerospace Systems & Management With Merit

2024

- Graduated with a 3.69 GPA
- Director's List (AY 2023/2024 Sem 1)
- MOE Edusave Merit Bursary

2023

- Executive member of NYP Women's Dragon Boat Team (2023-2024)

2022

- Director's List (AY 2021/2022 Sem 2)

2021

- Executive member of NYP Women's Dragon Boat Team (2021-2022)
- Director's List (AY 2021/2022 Sem 1)

LOYANG VIEW SECONDARY SCHOOL (2017-2020)

GCE O level

2020

- MOE Edusave EAGLES Award
- MOE Edusave Character Award
- National Youth Achievement Award Silver

2019

- President of Student Council (2019-2020)

2018

- Vice-Captain of Netball (2018-2019)
- National Youth Achievement Award Bronze
- MOE Edusave Scholarship

2017

- MOE Edusave Merit Bursary

Top Projects

HERMES (2026)

SUTD, 30.007 Engineering Design Innovation

- Partnered with Ento Industries to develop HERMES, a low-cost automated system for counting Black Soldier Fly neonates. Led the design of a deformable funnel, vibration-tuned ramp, and event-based camera vision pipeline, achieving 98.6% counting accuracy and 100% neonate survival at a cost of S\$1,120 per unit, under 1% of the commercial alternative.

SensOri (2026)

SUTD, 3D Design Project

- Designed and built a sensor-based system to help caregivers plan outings with greater confidence by enabling real-time monitoring and prediction of environmental conditions. The system collects and stores data on key sensory variables, sound levels, crowd density, and light intensity, using an ESP32 integrated with radar, sound, and light sensors.

AURAPLANE (2025)

SUTD, 03.007 Design Thinking & Innovation

- Collaborated in a team to design modular, multifunctioning partitions for both group and solo diners. Explored user journeys and conducted surveys to understand dining behaviors and preferences. Developed design solutions that enhanced privacy, flexibility and overall user experience.

Solar-Powered Doorbell (2025)

SUTD, 10.023 Designing Energy Systems

- Converted a doorbell to a fully solar-powered system while maintaining a functional and aesthetically appealing design. Performed calculations to select appropriate solar panels and battery capacity, optimizing energy efficiency and performance.

Parachute Delivery System Design (2024)

SUTD, 10.018 Design Thinking Project 1

- Designed and simulated a parachute-based payload delivery system using CAD (Rhino) and physics modelling. Tested the system using an IMU to measure acceleration and velocity, then compared the results with theoretical predictions. Used this analysis to improve stability and reduce impact in constrained deployment conditions.

Aircraft Fail-Safe DC Power Supply System (2022)

Nanyang Polytechnic

- Designed an aircraft fail-safe DC power supply system from scratch using only a schematic diagram. Planned and optimized component placement on a predefined circuit board size. Assembled and soldered all components onto the circuit board, ensuring system integrity